Introduction to Python Programming

1.

F01(▲):

● = 3

■ = ▲ + 2

★ = ■ - ●

return ★

SOLVE: F01(4)

■ =

★ =

● =

▲ =

ANSWER: ______

```
assert
```

AssertionError

pytest

```
pip install
pytest
```

docs.pytest.org

Fuel gauges indicate, often with fractions, just how much fuel is in a tank. For instance 1/4 indicates that a tank is 25% full, 1/2 indicates that a tank is 50% full, and 3/4 indicates that a tank is 75% full.

In a file called fuel.py, implement a program that prompts the user for a fraction, formatted as X/Y, wherein each of X and Y is an integer, and then outputs, as a percentage rounded to the nearest integer, how much fuel is in the tank. If, though, 1% or less remains, output E instead to indicate that the tank is essentially empty. And if 99% or more remains, output F instead to indicate that the tank is essentially full.

If, though, X or Y is not an integer, X is greater than Y, or Y is 0, instead prompt the user again. (It is not necessary for Y to be 4.) Be sure to catch any exceptions like ValueError or ZeroDivisionError.

The convert function expects a str in X/Y format as input, wherein each of X and Y is an integer, and returns that fraction as a percentage rounded to the nearest int between 0 and 100, inclusive. If X and/or Y is not an integer, or if X is greater than Y, then convert should raise a ValueError. If Y is 0, then convert should raise a ZeroDivisionError.

- gauge expects an int and returns a str that is:

  "E" if that int is less than or equal to 1,

  "F" if that int is greater than or equal to 99,

  and "Z%" otherwise, wherein Z is that same int.

```
def main():
    ...
def convert(fraction):
    ...
def gauge(percentage):
    …
if __name__ == "__main__":
    main()
```

Homework:
Then, in a file called test_fuel.py, implement two or more functions that collectively test your implementations of convert and gauge thoroughly, each of whose names should begin with test_ so that you can execute your tests with: